1. Clinical Question & Core Hypothesis

Clinical Question

What molecular mechanisms determine whether immune structures translate into clinical benefit?

Working Hypothesis

In pretreatment bladder urothelial carcinomas treated with cisplatin-based neoadjuvant chemotherapy, tumors that harbor dense, mature TLS-like lymphoid aggregates in direct proximity to tumor epithelium have a higher probability of therapy_response (defined as pathologic complete response or downstaging to ≤ pT1) after NAC. Tumors with sparse or immature TLS, or TLS located distal from the tumor front, show diminished or absent TLS-driven local immune activation and thus reduced therapy_response. Anchor_genes: [] (no Tier 1 anchor genes available). Secondary modifiers such as tumor–stroma ratio (TSR), tumor budding, overall intratumoral lymphocyte density, TLS maturation markers, and TLS–tumor proximity modulate this relationship but are not required anchors. Experimental approach (summary): quantify pretreatment TLS density, maturation, and spatial relationship to tumor epithelium via multiplex immunohistochemistry and spatial analyses; correlate with post-NAC pathology in a prospective cohort; adjust for clinical confounders; explore mechanistic proxies (cytotoxic T cell activation, ICD biomarkers) to support the proposed synergy.

Element	Value
Imaging feature	TLS-like lymphoid aggregates
Primary endpoint	therapy_response

2. Why This Gene — Not Others

Alternative / secondary genes considered

Tier	Gene	Role / Reason excluded from core
Secondary	Tumor–stroma ratio (TSR)	Supporting modifier from knowledge graph
Secondary	Tumor budding / invasive edge morphology	Supporting modifier from knowledge graph
Secondary	Overall lymphocyte density (tumor vs stroma)	Supporting modifier from knowledge graph

Secondary	TLS size/maturation (germinal-center-like features)	Supporting modifier from knowledge graph
Secondary	Proximity of TLS to invasive tumor front	Supporting modifier from knowledge graph

3. Clinically Testable Predictions

Prediction 1: At the tissue level, tumors with the imaging feature but low anchor gene activity will show immune cells confined to structural peripheries rather than infiltrating tumor nests.

4. Minimum Viable Validation Plan

A. Retrospective Cohort Analysis

Data source: TCGA-BLCA (Firehose Legacy), ~400 patients - Cox PH model with interaction term for the imaging feature and anchor gene - Kaplan-Meier 2x2 stratification

B. Tissue-Level Validation

Methods: Multiplex IF / IHC / spatial transcriptomics

Key readouts: - Immune structure markers (e.g., CD20, CD3, PNAd) - Effector cell localization (CD8) relative to tumor nests - Anchor gene protein / phospho-protein expression - Stromal features (collagen density by SHG, if applicable)

Goal: Confirm that the proposed mechanism operates at the spatial / tissue level.

C. Functional Validation

Models: Ex vivo organotypic cultures, 3D co-culture, syngeneic mouse models

Perturbations: Genetic (KO/rescue) or pharmacologic modulation of anchor gene

Readouts: T-cell infiltration, cytotoxicity (granzyme B / IFN-gamma), tumor growth / survival

Goal: Establish causal relationship between anchor gene activity and immune efficacy.

5. Strengths, Caveats & Alternative Directions

Strengths

- Novel mechanistic hypothesis derived from knowledge graph analysis.

Caveats & Risks

- Hypothesis is computationally derived and requires experimental validation.

Alternative Directions

- Tumor–stroma ratio (TSR): Could serve as an alternative mechanistic axis if the primary hypothesis is not supported.
- Tumor budding / invasive edge morphology: Could serve as an alternative mechanistic axis if the primary hypothesis is not supported.